

TECHNICAL INFORMATION

PTS-THERMOFLEX-60A23*9000 black

TPS, 58 Shore A

2024-06-27

Physical properties	MD / TD	Unit	Test Standard
Density	1.1	g/cm ³	ISO 1183
Molding shrinkage, test-plate 150 x 100 x 2 mm	1.8 / 1	%	Teknor Apex
Molding shrinkage depends on part geometry, wall thickness and processing conditions. The listed shrinkage values are not directly applicable for molded parts.			
Mechanical properties (TPE)	MD / TD	Unit	Test Standard
Hardness, Shore A, 3 s	58	Shore A	ISO 868
Tensile strength TPE	5 / 5	MPa	ISO 527-1/-2
Strain at break TPE	525 / 600	%	ISO 527-1/-2
Stress at x% strain			ISO 527-1/-2
10%	0.6 / 0.4	MPa	
50%	1.3 / 1.1	MPa	
100%	1.8 / 1.5	MPa	
300%	3.4 / 3.1	MPa	
Tear strength	14 / 15	kN/m	ISO 34-1 B
Compression set 1 x 6 mm			ISO 815-1
70 h 23 °C	14	%	
24 h 70 °C	30	%	
24 h 100 °C	50	%	
24 h 120 °C	66	%	
24 h 140 °C	81	%	
Thermal properties	MD / TD	Unit	Test Standard
Burning behaviour			UL94
1,6 mm	HB	-	
2 mm	HB	-	
3,2 mm	HB	-	
After-shrinkage/hot air (test-plate 150 x 100 x 2 mm)			Teknor Apex
24 h 70 °C	0.2 / 0.3	%	
24 h 100 °C	0.4 / 0.3	%	
24 h 120 °C	0.8 / 0.4	%	
Processing recommendation injection molding	Value	Unit	Test Standard
Melt temperature			-
Melt temperature	230 - 260	°C	
cold insert	270 - 280	°C	
Mold temperature	30	°C	-
Feed temperature	70	°C	-

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Zone 1	220	°C	-
Zone 2	230	°C	-
Zone 3	240	°C	-
Nozzle temperature	240	°C	-
Injection speed	high	mm/s	-
Maximum residence time			-
Maximum residence time	10	min	
cold insert	5	min	
Metering path - relation (L/D) of metering path length (L) to screw diameter (D)	1 - 3	D	-

Spiral flow	Value	Unit	Test Standard
Processing temperature	240	°C	-
Mold temperature	30	°C	-
Injection time	1	s	-
Spiral, width	5	mm	-
Spiral, thickness	1.5	mm	-
Length at injection pressure 400 bar	330	mm	-
Length at injection pressure 600 bar	450	mm	-
Length at injection pressure 800 bar	565	mm	-
Length at injection pressure 1000 bar	675	mm	-
Length at injection pressure 1200 bar	775	mm	-

Features

Features

high weld line strength, low compression set, Adhesion to PA, cold flexible

Processing

multi-component injection moulding, Injection Molding

Delivery form

Pellets

Applications

Automotive, Sealings, Electro, Electronic, Power tools, Domestic appliances, Household appliances

Product description

Further information:

The viscosity of TPEs is shear-dependent. Therefore injection speed and pressure show a much greater influence on flow behaviour than melt temperature alone.

Bonding will be improved with a humidity less than 0.05 %.

In order to avoid moisture uptake of the nylon before the softcomponent is injected, the delay between hard and soft component should be as short as possible.

To improve the bonding strength on cold inserts, the processing temperatures for cold inserts are recommended. For small parts and high residence times, slightly reduced cylinder temperatures are advantageous.

Material description according ISO 1043: TPS-P(PO)